

6

c

perfectly elastic collision :

$$v_Q - v_P = u_P - u_Q = 1.2 - (-1.8) = 3.0 \quad \text{-- Eqn 1}$$

COM :

$$(0.30)(1.2) - (0.60)(1.8) = 0.30 v_P + 0.60 v_Q \quad \text{-- Eqn 2}$$

Solve Eqn 1 and 2, $v_P = 2.8 \text{ m s}^{-1}$, $v_Q = 0.20 \text{ m s}^{-1}$

2

7 C Max pressure the probe can take = atmospheric pressure + pressure due